

# PhD MILESTONE 1 REPORT



## Economic Model Predictive Control of a Semi-Autogenous Grinding Mill Circuit: A Modular Energy-Minimising Cost Function

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# Contents

|                                                                                                                         |    |
|-------------------------------------------------------------------------------------------------------------------------|----|
| 1. Abstract . . . . .                                                                                                   | 2  |
| Nomenclature . . . . .                                                                                                  | 3  |
| 2. Research Background . . . . .                                                                                        | 4  |
| 2.1 The SAG grinding circuit and its control problem . . . . .                                                          | 4  |
| 2.2 Control-oriented dynamic models . . . . .                                                                           | 4  |
| 2.3 Disturbance observers and state estimation . . . . .                                                                | 7  |
| 2.4 The supervisory-optimisation continuum: SSRTO $\rightarrow$ ROPA $\rightarrow$<br>DRTO $\rightarrow$ EMPC . . . . . | 8  |
| 2.5 EMPC on grinding circuits - the empirical record . . . . .                                                          | 8  |
| 3. Research Gap and Problem Statement . . . . .                                                                         | 9  |
| 4. Aims and Objectives . . . . .                                                                                        | 11 |
| 5. Significance and Novelty . . . . .                                                                                   | 12 |
| 5.1 Theoretical contributions . . . . .                                                                                 | 12 |
| 5.2 Practical contributions . . . . .                                                                                   | 12 |
| 6. Research Methodology . . . . .                                                                                       | 13 |
| 6.1 The single-stage SAG grinding mill circuit . . . . .                                                                | 13 |
| 6.2 Control architecture . . . . .                                                                                      | 14 |
| 6.3 Forward work plan . . . . .                                                                                         | 16 |
| 7. Project Timeline . . . . .                                                                                           | 17 |
| 8. Ethics, Permits and Clearances . . . . .                                                                             | 18 |
| 9. Data Management . . . . .                                                                                            | 19 |
| 10. Resources and Budget . . . . .                                                                                      | 19 |
| 11. Conclusion . . . . .                                                                                                | 20 |
| References . . . . .                                                                                                    | 21 |

# 1. Abstract

Semi-autogenous grinding (SAG) is the most energy-intensive stage of mineral comminution. The comminution circuit dominates a concentrator’s electricity bill and the greenhouse emissions that come with it. The mill should grind the ore to the size the downstream circuits require, at the lowest possible energy per tonne, while respecting throughput demand and product-quality limits. The classical solution places a steady-state real-time optimiser (SSRTO) above a tracking model predictive controller (MPC). Ore hardness and feed size drift on hour-to-shift timescales, so the plant rarely holds the strict steady state that SSRTO’s parameter-estimation step requires. The optimiser then sits idle and pushes stale set-points. Economic MPC (EMPC) merges optimisation and regulation into a single receding-horizon problem. It can track a moving economic optimum during transients, and transients are where a SAG mill spends most of its operating life.

This research develops and benchmarks an EMPC framework for a SAG circuit. It uses two prediction models. The Hulbert-Le Roux reduced-order model handles fast online solves. A cumulative-rates population balance model (PBM) covers cases that need the full product size distribution. Two contributions are central. The first is a modular, energy-minimising cost function. The online controller minimises mill energy alone. The downstream economics do not enter it directly. They are worked out offline and reduced to one specification, a particle-size floor the controller must hold. Re-tuning for a new ore body or a different downstream plant then means swapping that specification, not redesigning the controller. The second is a comparison of the supervisory architectures (SSRTO, ROPA, and single- and two-layer EMPC) together with the basic regulatory baselines. The test is not a fixed steady state but a plant-representative, transient-rich scenario.

Preliminary work has implemented all of these controllers in `do-mpc/CasADi/IPOPT` on a simulated eight-state circuit. The first like-for-like benchmark is now complete. Six 218 h closed-loop runs compared three architectures (SSRTO over PI loops, single-layer EMPC and two-layer EMPC) under both economic objectives, on identical ore, estimator and constraints. Two findings stand out. First, the specific energy of all six runs clustered within about two percent and followed the ore hardness. The energy per tonne is set by the ore and the quality floor, not by the architecture or the objective. Second, energy minimisation reached the same specific energy as profit maximisation while keeping the mill away from its feed and speed limits. This supports the modular mill-level objective the project proposes. The two-layer EMPC remains the main candidate. These results are synthetic. The next step is to calibrate the mill model against historical plant data, so the comparison can carry plant-level numbers.

## Nomenclature

**Manipulated variables (MV):** MFS mill feed solids / ore feed rate (t/h) · CFF cyclone feed flow · SFW sump feed water ·  $\phi_c$  mill speed (fraction of critical speed, 0.55-0.80).

**Controlled variables (CV):** JT total mill charge filling · SVOL sump slurry volume · PSE product size estimate (fraction < 75  $\mu$ m) · Pmill mill power draw.

**Plant states (8):** mill Xmw, Xms, Xmf, Xmr, Xmb (water, solids, fines, rocks, balls); sump Xsw, Xss, Xsf (water, solids, fines).

**Disturbances / parameters:**  $\alpha_r$  feed rocks fraction ·  $\phi_f$  ore-hardness (power per tonne of fines) · MIW mill inlet water · MFB ball feed.

**Derived / economic:** TP throughput ( $\approx$  MFS at steady state) · SEC =  $P_{mill}$ /MFS specific energy consumption (kWh/t) · A net return per tonne (\$/t) · B electricity tariff (\$/kWh) · NSR net smelter return ·  $J_{comm}$  commercial objective.

**Acronyms:** SAG semi-autogenous grinding · PBM population balance model · PSD particle-size distribution · (E)MPC (economic) model predictive control · RTO real-time optimisation · SSRTO steady-state RTO · ROPA RTO with persistent adaptation · DRTO dynamic RTO · EKF extended Kalman filter · DO/NDOB (nonlinear) disturbance observer · PINN physics-informed neural network · OCP optimal control problem · NLP nonlinear program.

## 2. Research Background

### 2.1 The SAG grinding circuit and its control problem

Comminution is the crushing and grinding of run-of-mine (ROM) ore to free the valuable mineral. It is the single largest electricity consumer on a concentrator, and the SAG mill is the largest single draw within it. The control problem is multivariable and strongly coupled. A closed circuit (SAG mill  $\rightarrow$  sump  $\rightarrow$  hydrocyclone) is driven through a handful of manipulated variables: ore feed rate, cyclone feed flow, dilution water and mill speed. The controller must hold a few controlled variables (mill charge, sump level, product particle size) inside an operating envelope. Meanwhile the ore properties drift underneath the controller. The economically meaningful quantity is the specific energy consumption (SEC), the energy per tonne of product at acceptable quality. SEC is also the right variable for comparing controllers, and it is the headline metric of this project’s benchmark. Set-point tracking remains necessary at the regulatory level. On its own, however, it does not decide where the plant should operate. Choosing the operating point is an economic decision.

A controller is only as good as the model behind it, the estimator that feeds it state, and the supervisory layer that decides *where* to operate. The next three subsections review each in turn.

### 2.2 Control-oriented dynamic models

A control-oriented mill model must be predictive enough to capture the load, power, and product dynamics. It must also stay cheap enough to solve inside an online optimisation. Two mechanistic families span this trade-off, and both are used in this work. A reduced-order lumped model serves the fast economic MPC. A size-resolved population balance model serves the cases where the cost function must see the full product size distribution.

The **Hulbert-Le Roux reduced-order model** (Le Roux et al., 2013; Le Roux & Steyn, 2022) collapses the size structure to three operationally meaningful lumps: *rocks*, *solids* and *fines*. Water and ball hold-ups are added, giving a five-state mill plus a three-state sump in the closed-circuit form. The model’s strengths make it the default for online control. It can be identified from a single plant survey. It supports non-linear MPC at industrial sample rates (Coetzee et al., 2010; Le Roux et al., 2016). An EKF state estimator works well on it, recovering the unmeasured internal states from standard plant sensors (Le Roux et al., 2017). The model has also been embedded in an industrial digital twin (Quintanilla et al., 2025). The model’s drawbacks follow from the same lumping, the grouping of all particle sizes

into three bins. The single quality output is the product particle-size estimate (PSE), the fraction of cyclone-overflow product passing a specified limit (e.g. 75  $\mu\text{m}$ ). The model therefore represents only one quantile of the size distribution and never its shape.

The model's core is a set of population volume balances for the mill hold-ups of water, solids, rocks and fines, in  $\text{m}^3$  (Le Roux et al., 2013):

$$\frac{d}{dt}X_{mw} = MIW + V_{cwu} - V_{mwo} \quad (1a)$$

$$\frac{d}{dt}X_{ms} = \frac{MFS}{D_S} (1 - \alpha_r) + V_{csu} - V_{mso} + RC \quad (1b)$$

$$\frac{d}{dt}X_{mr} = \frac{MFS}{D_S} \alpha_r - RC \quad (1c)$$

$$\frac{d}{dt}X_{mf} = \frac{MFS}{D_S} \alpha_f + V_{cfu} - V_{mfo} + FP \quad (1d)$$

Each hold-up gains material from the feed and the cyclone underflow and loses it through the end-discharge screen (the  $V$  flow terms). A fifth balance tracks the ball hold-up. The feed ore splits by two ore parameters. The fraction  $\alpha_r$  enters as rocks, too large to discharge. The fraction  $\alpha_f$  enters as fines, already below the specification size. Breakage moves material down this ladder, and in this model it is driven by mill power:

$$RC = \frac{P_{mill} \varphi}{D_S \phi_r} \cdot \frac{X_{mr}}{X_{mr} + X_{ms}} \quad (2a)$$

$$FP = \frac{P_{mill}}{D_S \phi_f \left[ 1 + \alpha_{\phi_f} \left( \frac{LOAD}{v_{mill}} - v_{P_{max}} \right) \right]} \quad (2b)$$

where  $RC$  is the volumetric rate at which rocks are consumed into solids and  $FP$  the rate at which new fines are produced. The two ore-hardness parameters sit exactly here.  $\phi_r$  (kWh/t) is the rock abrasion factor, the energy required per tonne of rocks broken.  $\phi_f$  (kWh/t) is the power needed per tonne of fines produced, the ore-hardness parameter used throughout this report. The factor  $\varphi$  is the slurry rheology factor, and  $\alpha_{\phi_f}$  is a small correction for the mill filling ( $LOAD$  is the total charge volume and  $v_{mill}$  the mill volume). Mill power itself is an empirical parabola in the charge and the rheology, scaled by the mill speed (Le Roux et al., 2013):

$$P_{mill} = P_{max} \{1 - \delta_{Pv} Z_x^2 - 2\chi_P \delta_{Pv} \delta_{Ps} Z_x Z_r - \delta_{Ps} Z_r^2\} (\alpha_{speed})^{\alpha_P} \quad (3)$$

where  $Z_x$  measures the charge volume against the filling at maximum power,  $Z_r$  measures the rheology factor against its value at maximum power, and  $\alpha_{speed}$  is the fraction of critical mill speed (written  $\phi_c$  elsewhere in this report).

The **cumulative-rates population balance model** resolves the full PSD. It tracks the ore mass in each of  $n$  geometrically-spaced size classes, giving a  $2n+2$ -state circuit ( $n$  mill classes plus mill water,  $n$  sump classes plus sump water). Its defining feature is the cumulative-rates parametrisation of Hinde & Kalala (2009). A standard PBM describes breakage with two functions that are hard to measure: a specific breakage-rate function and a breakage-distribution function. Their parameters are back-calculated from test data and are notoriously sensitive to measurement error, with large variances. The cumulative-rates form replaces both with a single cumulative breakage rate  $K_i^E$ . This is the rate at which material coarser than size  $x_i$  breaks below that size, normalised by mill power so that the rate is largely insensitive to scale-up. The decisive advantage of the cumulative-rates form is direct estimation.  $K_i^E$  comes straight from measured feed and product size distributions and the mill's specific energy input, with no back-calculation. That makes a PSD-resolving model robustly identifiable, and smooth enough to linearise or differentiate for control.

For size class  $i$  the model tracks the mass  $w_i$  held in the mill, with  $W_i$  the mass coarser than size  $x_i$  (Le Roux & Craig, 2013):

$$\frac{dw_i}{dt} = f_i - g_i w_i + W_i \frac{P}{M} K_i^E - (w_i + W_i) \frac{P}{M} K_{i+1}^E \quad (4)$$

where  $f_i$  is the feed rate into the class,  $g_i$  its discharge rate,  $P$  the net mill power and  $M$  the ore hold-up. The first power term is material breaking into the class from coarser sizes. The second is material breaking out of it to finer sizes. The energy-normalised rate  $K_i^E = (M/P) K_i$ , in  $(\text{kWh/t})^{-1}$ , is the scale-robust quantity described above, and it is where the ore hardness lives in this model. Harder ore means smaller  $K_i^E$  values, fewer tonnes broken below each size per kWh of grinding energy. The whole rate curve is captured by one smooth fitted function (Le Roux & Craig, 2013):

$$\hat{K}_i^E = \kappa_1 \frac{x_i^{\alpha_1}}{1 + (x_i/\mu)^\lambda} + \kappa_2 x_i^{\alpha_2} \quad (5)$$

with six ore-dependent parameters  $\kappa_1$ ,  $\kappa_2$ ,  $\alpha_1$ ,  $\alpha_2$ ,  $\mu$  and  $\lambda$ . Together they play

the role that the single  $\phi_f$  plays in the reduced-order model, spread over the full size axis.

The two families are complementary rather than competing. The Hulbert-Le Roux model is preferred for online MPC. Its small state keeps the optimisation cheap and offers the right granularity for grinding-circuit-only (PSE-based) economic decisions. The cumulative-rates PBM is needed when the cost function must see PSD shape, for example when downstream flotation and leaching kinetics are size-dependent. Any genuinely plant-wide economic objective requires a PSD-resolving signal that a single PSE quantile cannot provide. The reduced  $n = 5$  or  $9$  form supplies that at a state count a controller can still afford.

## 2.3 Disturbance observers and state estimation

A SAG circuit’s disturbances are well characterised and act on distinct timescales. Ore-hardness drift acts hour-to-shift and is the dominant source of model-plant mismatch. Feed-size variability acts on minutes-to-hours and is usually correlated with hardness. Feed-moisture and dilution-water swings change the slurry rheology. Liner wear and ball-charge drift act over weeks to months. The literature splits the estimation work across two roles. A fast non-linear disturbance observer (NDOB) at the regulatory MPC layer cancels input-side disturbances by feed-forward before they reach the optimisation loop. The first grinding-circuit deployments (Chen et al., 2009; Yang et al., 2010) established two things. Pairing a DO with the MPC improves disturbance rejection without changing the controller’s normal set-point tracking. The estimate also stays meaningful to operators, who can read it directly as an “ore-hardness offset”. The second role is a slow estimator at the economic layer, an augmented EKF/UKF (Le Roux et al., 2017). It tracks the drifting hardness parameter and recovers the unmeasured internal mill states. The augmented Kalman filter is the established choice for this role on grinding circuits. It propagates the slow parameter drift through its process-noise model while staying a deterministic, real-time update. Here the augmented estimator tracks a single slow parameter, the ore hardness  $\phi_f$  (the power drawn per tonne of fines produced). One direction this work will explore is augmenting it with further slowly-varying disturbance states, such as the feed rocks fraction  $\alpha_r$  and other rheology-related parameters. The test is whether jointly estimating them, rather than  $\phi_f$  alone, sharpens the economic-layer model under the realistic, multi-disturbance scenario.



## 2.4 The supervisory-optimisation continuum: SSRTO $\rightarrow$ ROPA $\rightarrow$ DRTO $\rightarrow$ EMPC

The question “*how does the controller decide what the economically best operating point is?*” has four published answers. **Classical SSRTO** (Darby et al., 2011) puts a separate steady-state optimiser above the MPC. It solves a parameter-estimation and economic optimisation problem whenever a steady-state detector declares the plant settled. Between detections the set-points are held. The wait is necessary. Fitting the steady-state model to transient data corrupts it. For a SAG mill this is a real problem. The plant rarely holds the strict steady state the detector requires, so the optimiser sits idle and pushes stale set-points.

**ROPA** (real-time optimisation with persistent adaptation) breaks SSRTO’s central coupling. It runs a continuous dynamic estimator for the model parameters, so the still-steady-state economic NLP always receives fresh parameters, even during transients. **DRTO** replaces the static NLP with a dynamic optimisation over a long horizon. It emits an economic *trajectory* for a tracking MPC to follow. **EMPC** (Ellis et al., 2014; Faulwasser & Pannocchia, 2018) takes the DRTO logic to its limit. It drops the dedicated upper layer and embeds the economic objective directly in the regulatory MPC’s stage cost. DRTO and EMPC are theoretically equivalent. They share the same dynamic economic cost, model and horizon, and differ only in whether the layers are kept visibly separate. An experimental-rig comparison (Matias et al., 2022) reports ROPA matching DRTO at about 38 % profit improvement over a fixed-input baseline, while SSRTO reaches only about 20 %. The practical ordering is  $\text{EMPC} \approx \text{DRTO} > \text{ROPA} \gg \text{SSRTO}$ .

## 2.5 EMPC on grinding circuits - the empirical record

The components have been published piecewise. Thivierge, Bouchard and Desbiens (2023) provide the closest precedent. They compare four controllers on a simulated HPGR, ball mill and flotation circuit: basic regulatory control (BRC), advanced regulatory control (ARC), a profit EMPC, and a power-penalised EMPC. The economic cost function is flotation-concentrate revenue minus the energy cost. Three of their results matter here. The basic regulatory baseline used significantly more energy than the optimising controllers. The ARC and the profit EMPC performed identically, because the economic optimum sits at the process constraints and both end up holding the plant there. The power-penalised EMPC cut specific energy by only 0.4 to 2.9 percent relative to the profit optimum, and gave up 3.6 to 3.9 percent of net value for it. However, they evaluate only at steady-state operating points after the simulator re-settles, and defer transient performance to future work. The transient-rich benchmark of this project takes exactly that missing

step, and where the two studies overlap, our first six-run results agree with theirs. Numbi & Xia (2016) demonstrate energy-cost EMPC on a crushing plant under time-of-use tariffs. Jia et al. (2020) demonstrate multi-stage EMPC for downstream gold cyanidation. No single study combines the Hulbert-Le Roux model, a composite DO + augmented-EKF estimator stack, and an explicit specific-energy EMPC cost on a SAG circuit, evaluated under realistic operating conditions.

### 3. Research Gap and Problem Statement

The literature reviewed in Section 2 traces a body of work in which every component of an EMPC for a SAG circuit has been published. They have never been assembled into a single closed-loop study. They have also never been exercised under the conditions an industrial SAG mill actually experiences. Three connected gaps follow.

**Gap 1 - Architectural integration on a SAG circuit.** No published study integrates a regulatory-layer DO, an upper-layer augmented EKF/UKF, and an economic stage cost into a single closed-loop SAG-mill controller. The two estimator approaches address complementary timescales but have evolved independently. DO + MPC cancels fast input-side disturbances at the regulatory clock. The augmented-KF approach absorbs slow parameter drift at the economic clock.

**Gap 2 - Realistic operating conditions in evaluation.** Industrial SAG mills settle at *operational* steady state often, but the underlying conditions never do. Ore hardness drifts hour-to-shift. Liner wear changes the mill geometry over weeks to months. The plant therefore moves between mildly suboptimal operating points rather than running at the economic optimum. Published grinding-circuit studies report performance under tightly controlled conditions instead: step disturbances at fixed operating points, with metrics sampled after re-settling. Two things have not been quantified. The first is how a SAG-mill EMPC behaves under disturbance sequences that resemble actual plant operation. The second is how large the economic opportunity is that EMPC could capture by tracking the moving optimum during transients. This quantification is the prerequisite for any fair head-to-head comparison of EMPC against SSRTO, ROPA, or DRTO.

**Gap 3 - Cost-function scope on a SAG circuit.** The plant-wide framework of Le Roux & Craig (2019) and the flotation-revenue cost of Thivierge (2023) both argue that the mill’s economic objective should pull from downstream plant signals. Most grinding-circuit EMPC studies accordingly adopt a revenue- or profit-based stage cost. Two problems with that choice on a single-stage SAG mill have gone unaddressed. The first is a scale mismatch. A profit term such as

$A(\text{PSE}) \text{ TP} - B P_{\text{mill}}$  prices the product through a per-tonne return  $A(\text{PSE})$ . That one number collapses recovery, grade, metal price and downstream constraints. All of these belong to the plant. The mill controller cannot measure or control them. Embedding these downstream economics in the mill controller couples it to a calibrated flotation model, a specific ore body and a specific market state. The controller must then be re-identified whenever any of these change. The second problem is a trivial optimum. Even with  $A$  known, revenue scales with throughput while energy is a small per-tonne penalty. Maximising the profit form therefore simply drives mill speed  $\phi_c$  and feed **MFS** to their limits. The mill runs flat out against its throughput and overload constraints. That is a maximum mode that needs no optimiser, not a meaningful interior optimum.

A complementary and unexplored direction resolves both problems: lift the revenue term out of the mill controller entirely. The plant-wide layer sets the product-size floor and the throughput demand. The mill controller minimises mill energy for those targets. The stage cost is composed from a small, extensible library of mill-owned terms. The library holds mill energy ( $B P_{\text{mill}}$ ), specific energy consumption (SEC, energy per tonne of feed ore), and penalties on mill-specific quantities such as residence time. With a size-resolved PBM it can also penalise the coarse tail of the product PSD. Terms are used singly or in combination, and weighted, as the downstream specification requires.

The two levels meet at a two-number interface. The first number is the throughput demand, a floor on **MFS**. The plant-wide optimisation, which sees all circuits, decides it and hands it down. The second is the product-size floor, set by the downstream circuits' requirements. Inside those limits the mill runs autonomously and never needs to maximise **MFS** itself. Re-targeting the mill is a change of two numbers rather than a re-identification. Neither the modular framing nor any of these mill-local objectives has yet been instantiated as the closed-loop cost function of a single-stage SAG mill.

These three gaps define the problem this thesis addresses: to build an integrated single-stage SAG-mill EMPC with a composite DO + augmented-estimator stack, to give it a modular energy-minimising cost function, and to benchmark it honestly against the full RTO/EMPC architecture spectrum under operating conditions representative of a real plant.

## 4. Aims and Objectives

The aim of this research is to develop and rigorously evaluate an economic model predictive control framework for a single-stage SAG grinding circuit that minimises specific energy consumption under realistic operating conditions, and to establish where it offers a genuine advantage over the established real-time-optimisation architectures. This aim is pursued through four objectives.

1. **Implement and validate the integrated control stack.** Build the single-stage SAG controller in `do-mpc`/CasADi/IPOPT and verify closed-loop regulation against the simulated plant. The controller runs on both the Hulbert-Le Roux reduced-order model and the cumulative-rates PBM, used in parallel according to the strengths and drawbacks of each (Section 2.2). Estimation uses the EKF / augmented EKF plus a fast disturbance observer. (*Substantially complete.*)
2. **Formulate a modular, energy-minimising EMPC cost function.** Derive and implement a stage cost in which the downstream plant economics enter only through an offline-derived particle-size specification (a one-sided PSE/PSD floor). The online controller then minimises mill energy at the demanded throughput and quality. Demonstrate the modularity claim: adapting to a new ore body or downstream topology is a swap of the specification interface, not a controller redesign (the interface is detailed in Section 6.2). Explore modular cost functions suited to both single- and multi-stage SAG circuits, where each mill’s local objective couples to the next only through the specification interface. (*Primary contribution.*)
3. **Construct a realistic transient scenario and benchmark the architectures.** Build a plant-representative, transient-rich operating scenario calibrated to historical plant data. It combines ore-hardness drift, feed-size variability, and operating-point changes that drive the plant between steady states. Use it to compare SSRTO, ROPA, and single- and two-layer EMPC against a decentralised-PI / tracking-MPC baseline, on specific energy consumption and constraint compliance. (*Primary contribution.*)
4. **Develop a PINN surrogate for online tractability.** Train a physics-informed neural-network surrogate offline on PBM-simulated data. Substitute it for the mechanistic prediction model inside the EMPC. The goal is to keep the online optimisation tractable as model fidelity and horizon length grow.

**Scope and priority.** Objectives 1-3 form the core of the candidature and are largely instantiated in the preliminary work. **Objectives 2 and 3 are the cen-**

**tral novel contributions** (the modular cost function and the transient benchmark). Objective 4 (PINN) is optional. It is pursued only if online solves become too slow as the horizon and model detail grow. Otherwise the mechanistic model stays the default. This staging keeps the thesis deliverable even if Objective 4 yields negative or partial results.

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## 5. Significance and Novelty

### 5.1 Theoretical contributions

This work would be the first to bring three pieces together inside a single closed-loop SAG-mill controller: a regulatory-layer disturbance observer, an upper-layer augmented EKF, and an explicit specific-energy economic stage cost. That closes the architectural gap the component literature leaves open (Gap 1). The modular spec-interface cost function (Objective 2) is a novel framing of the plant-wide economic objective. It separates the offline downstream-economics question from the online control problem. The transient-scenario benchmark (Objective 3) provides the first quantification on a SAG mill of the economic opportunity EMPC captures during transients. That measurement is what any honest EMPC-vs-RTO comparison requires.

### 5.2 Practical contributions

The outcome is a portable, energy-minimising SAG-mill controller. Its economic objective can be retargeted to a different ore body or concentrator configuration by replacing an offline-derived specification, not by re-engineering the controller. That lowers the barrier to industrial reuse. The transient-scenario study yields credible, plant-representative specific-energy-saving figures rather than figures from a benign synthetic test. Practitioners gain a defensible basis for deciding between an RTO retrofit and a full EMPC.

EMPC has clear advantages over classical RTO, yet it is rarely used in operating plants. Two practical barriers explain much of the gap. The first is the well-documented bang-bang issue. When the quality floor is already satisfied, the energy cost is nearly flat in feed. With no clear optimum, the controller swings between its actuator bounds. The second barrier is complexity at the plant-wide level. There, economic calculations at the plant-wide level run together with planning and scheduling information and with the dynamics of many units, so steady-state calculations remain the practical choice. This project reduces both barriers. Modularity removes part of the plant-wide complexity. The plant-wide

layer hands the mill its required interface values for variables such as PSE and TP, instead of implementing the mill inside its own optimisation. The two-layer EMPC at the SAG mill reduces the volatile bang-bang issue significantly using the longer horizon at the upper layer. It also separates the economic decision from the moment-to-moment actuation. Using complex cost functions, such as a moving average of SEC, has the potential to remove the volatile swings completely.

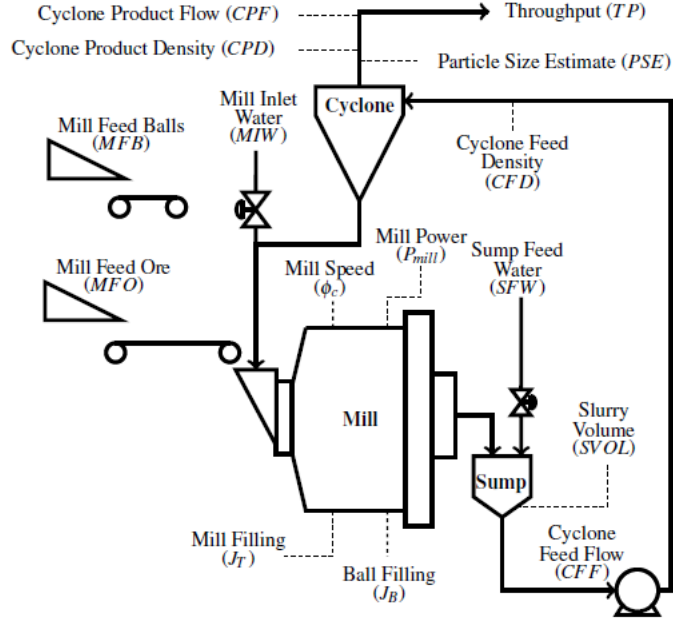
Comminution dominates concentrator electricity use. Grinding alone can draw up to roughly 80 % of a concentrator’s energy. Even single-digit-percent specific-energy reductions therefore translate into material cost and emissions savings at industrial scale. By cutting grinding electricity demand and the associated greenhouse emissions, the project supports Western Australia’s Net Zero by 2050 commitment. It also aligns with Curtin University’s research direction in sustainable mineral processing.

## **6. Research Methodology**

The methodology is organised in three parts: the process under study and its control problem (Section 6.1), the control architecture proposed for it (Section 6.2), and the forward work plan that delivers and evaluates the contributions phase by phase against the objectives of Section 4 (Section 6.3).

### **6.1 The single-stage SAG grinding mill circuit**

The case study throughout is a single-stage grinding mill circuit closed by a hydrocyclone (Figure 1). This is the configuration most commonly used for single-stage grinding and the standard testbed for grinding-circuit control (le Roux et al., 2013). Its three elements are a SAG mill, a sump, and a cyclone.



**Figure 1.** The single-stage SAG grinding mill circuit (after le Roux et al., 2013): SAG mill, sump, and cyclone, with the manipulated variables (MFS, CFF, SFW,  $\phi_c$ ), the controlled variables (JT, SVOL, PSE,  $P_{mill}$ ), and the circuit outputs (throughput TP, cyclone-feed density CFD, cyclone product flow CPF and density CPD).

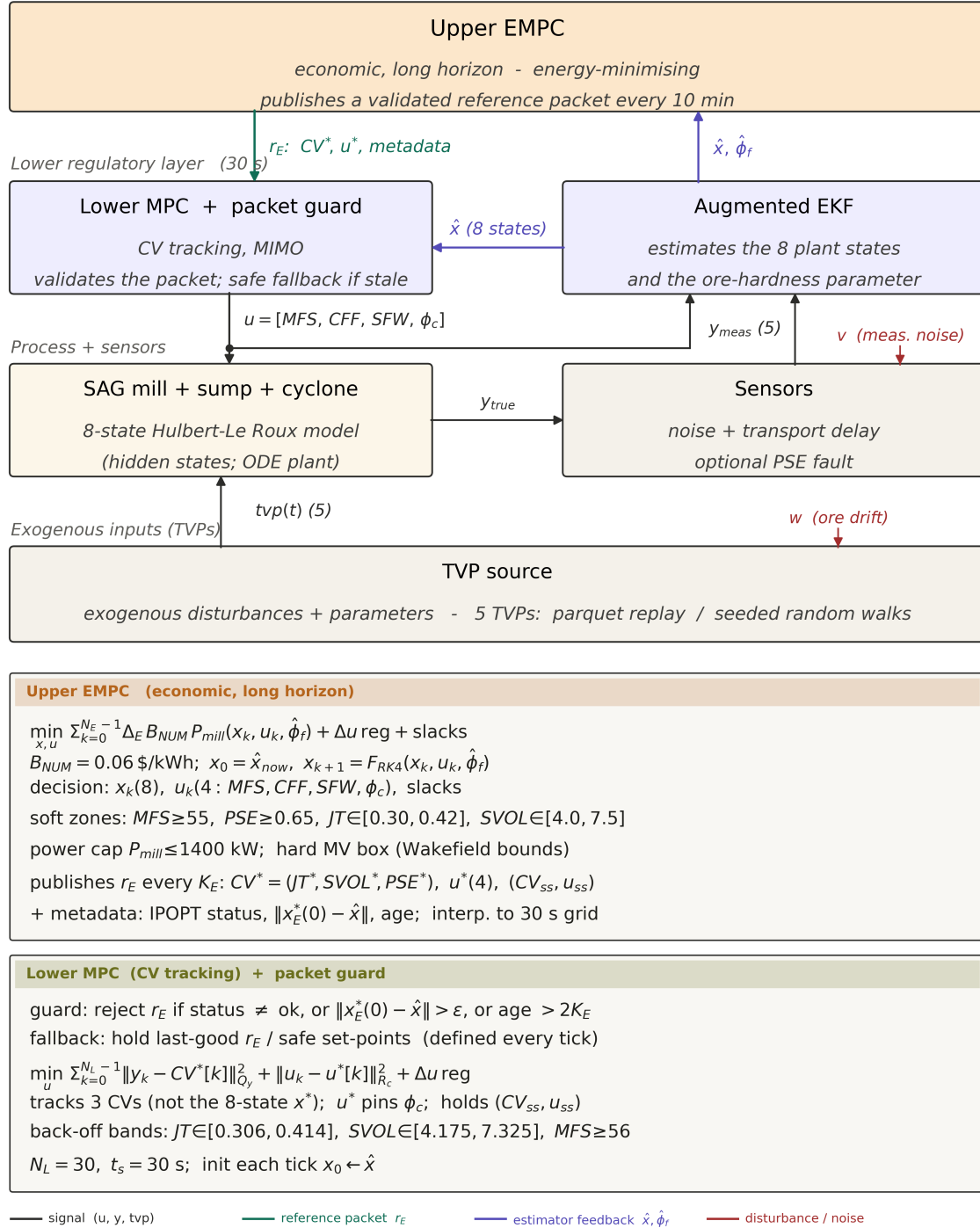
Run-of-mine ore (MFS), steel balls (MFB), and inlet water (MIW) are fed to the mill. The mill is driven at a fraction of critical speed ( $\phi_c$ ). It grinds the charge and discharges slurry to the sump. Dilution water (SFW) sets the density there, and the cyclone feed pump delivers the slurry at flow CFF to the cyclone. The cyclone returns coarse underflow to the mill and sends fine overflow out as product. The manipulated variables are therefore the ore feed rate MFS, the cyclone feed flow CFF, the sump dilution water SFW, and the mill speed  $\phi_c$ . The controlled variables are the mill charge filling JT, the sump slurry volume SVOL, and the product particle-size estimate PSE (the fraction of overflow finer than a target mesh). Mill power  $P_{mill}$  is monitored as a load and overload indicator. Throughput TP (the product mass rate, approximately MFS at steady state) and PSE are the circuit’s economic outputs.

## 6.2 Control architecture

The circuit runs under disturbances on distinct timescales. Ore hardness ( $\phi_f$ ) drifts hour-to-shift and is the dominant source of model-plant mismatch. Feed size and the feed rocks fraction ( $\alpha_r$ ) vary over minutes to hours. Liner wear and ball-charge drift act more slowly. The control objective is economic: grind to the required size at the demanded throughput for the least specific energy, while holding the mill inside its safe operating envelope.

## Two-layer EMPC control architecture (SAG mill circuit)

Upper EMPC publishes a validated reference packet; the lower MPC tracks it; an augmented EKF feeds both layers.  
Upper economic layer (10 min, 4 h horizon)



**Figure 2.** The two-layer EMPC control architecture: the upper EMPC (economic, long horizon) publishes a validated reference packet ( $CV^*$ ,  $u^*$ ) through a packet guard to the lower CV-tracking MIMO MPC; an augmented EKF estimates the mill and sump states plus the ore-hardness parameter, and a disturbance/parameter (TVP) source drives the simulated plant.



Figure 2 shows the two-layer EMPC realisation. At the regulatory clock, a fast non-linear disturbance observer cancels input-side disturbances and a lower MIMO MPC tracks references. At the economic clock, an upper EMPC computes the economically-optimal operating policy over a long horizon, with all four MVs as decision variables. The upper layer does not drive the plant directly. It publishes a reference packet, the CV and MV trajectories  $\mathbf{CV}^*$  and  $\mathbf{u}^*$ . A packet guard validates each packet on solver status, initial-state residual, and age before the lower MPC tracks it. When a solve is rejected, the guard falls back to the last valid packet or a safe set-point. An augmented extended Kalman filter estimates the eight mill and sump states together with the slow ore-hardness parameter  $\phi_f$ , and feeds both layers.

The economic objective enters the upper EMPC as the modular, mill-local stage cost of Objective 2. The online cost minimises a composable mill-owned term (mill energy, specific energy consumption, or another term from the library of Section 3) subject to the offline-derived particle-size floor. The prediction step uses the two model families of Section 2.2, the Hulbert-Le Roux model for fast online solves and the cumulative-rates PBM where the cost must resolve the product size distribution.

### 6.3 Forward work plan

The remaining programme is organised into four phases mapped to the objectives of Section 4. Each phase states its method and deliverable.

**Phase A - Mill-model calibration and transient-scenario construction (Objectives 1 and 3).** *Method.* Obtain historical operating data for a SAG circuit under a data-sharing agreement (Section 8). Fit the Hulbert-Le Roux and cumulative-rates parameters to it. Distinguish identifiable parameters from the historical data, and validate against held-out operating periods. From the same data, construct a plant-representative input sequence. The sequence combines ore-hardness drift and feed variability, calibrated in amplitude and timescale. It adds operating-point changes that drive the plant between steady states. These transitions are where the supervisory architectures are expected to separate. Characterise the sequence by the fraction of runtime spent in transient versus operational steady state.

*Deliverable.* A plant-calibrated mill model with a documented identifiability analysis, together with a reproducible transient scenario and its documented statistics.

**Phase B - Modular energy-minimising cost-function formulation (Objective 2).** *Method.* Formalise the specification interface. Compile the down-

stream economics offline into a per-tonne return  $\mathbf{A}$  and a one-sided PSE/PSD floor derived from downstream recovery and grade behaviour. Define the online stage cost as a composable, mill-local objective (energy, SEC, or other mill-owned terms) subject to that floor. Explore cost-function forms for both single- and multi-stage SAG circuits. Each mill optimises its own energy through the interface, so the same cost chains from one mill to the next. Each mill’s product-size floor is set by the following mill’s feed requirement.

*Deliverable.* A modular cost-function specification and its `do-mpc` implementation, with the offline-derivation procedure documented.

**Phase C - Architecture benchmark under the transient scenario (Objective 3).** *Method.* Run the full controller set on the Phase A scenario: decentralised PI, tracking MPC, single- and two-layer EMPC, SSRTO, and ROPA. Hold the estimator stack and the disturbance/sensor model fixed, so that only the supervisory architecture varies. Evaluate on specific energy, constraint compliance (PSE-floor violations), actuation smoothness, and the economic objective, with confidence intervals across scenario realisations.

*Deliverable.* A benchmark quantifying the specific-energy gap between architectures under transients.

**Phase D - PINN surrogate for online tractability (Objective 4).** *Method.* Train a physics-informed neural-network surrogate offline on PBM-simulated data, embedding the governing residual in the loss. Substitute it for the mechanistic prediction model inside the EMPC. Quantify the online speed-up and any accuracy or economic cost.

*Deliverable.* A PINN surrogate and a tractability/accuracy assessment within the EMPC loop.

## 7. Project Timeline

The candidature runs over three and a half years, from commencement on 1 March 2026 to thesis submission and defence in September 2029. The Gantt chart below resolves the phases of Section 6.3 to the month. It places the three candidacy milestones: the candidacy report at month 6, the mid-candidacy review at month 18, and the pre-submission review at month 39, three months before the defence. The first six months, now complete, cover the preliminary implementation of the integrated control stack and the Objective 1 baseline reported here. The programme from Phase A onward is the forward plan.

| Task                                         | Year | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec |
|----------------------------------------------|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Literature review                            | 2026 |     |     |     |     |     |     |     |     |     |     |     |     |
| Integrated control stack (preliminary)       |      |     |     |     |     |     |     |     |     |     |     |     |     |
| <b>Milestone 1: Candidacy (this report)</b>  |      |     |     |     |     |     |     |     |     | ◆   |     |     |     |
| Phase A: calibration + scenario (O1, O3)     |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Phase A: calibration + scenario (O1, O3)     | 2027 |     |     |     |     |     |     |     |     |     |     |     |     |
| Phase B: modular cost function (O2)          |      |     |     |     |     |     |     |     |     |     |     |     |     |
| <b>Milestone 2: Mid-Candidacy (Month 18)</b> |      |     |     |     |     |     |     |     |     | ◆   |     |     |     |
| Phase C: architecture benchmark (O3)         |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Phase C: architecture benchmark (O3)         | 2028 |     |     |     |     |     |     |     |     |     |     |     |     |
| Phase D: PINN surrogate (O4)                 |      |     |     |     |     |     |     |     |     |     |     |     |     |
| PINN-EMPC integration and tractability (O4)  |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Industrial implementation and interfacing    |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Industrial implementation and interfacing    | 2029 |     |     |     |     |     |     |     |     |     |     |     |     |
| Industrial validation and integration        |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Analysis, reporting and dissemination        |      |     |     |     |     |     |     |     |     |     |     |     |     |
| Thesis writing                               |      |     |     |     |     |     |     |     |     |     |     |     |     |
| <b>Milestone 3: Pre-Submission</b>           |      |     |     |     |     |     |     |     |     |     |     |     |     |
| <b>Thesis submission and defence</b>         |      |     |     |     |     |     |     |     |     |     |     |     |     |

## 8. Ethics, Permits and Clearances

This research does not involve human or animal participants, so formal ethics clearance is not required. The work is computational: simulation, control design, and data analysis. One non-trivial requirement does apply. The plant-calibration and transient-scenario phase (Phase A, Section 6.3) depends on historical operating data from an industrial SAG circuit. That will require a data-sharing agreement and, where applicable, a non-disclosure agreement with the providing operator. Confidential data will be handled, stored, and reported in accordance with the agreement terms and the Australian Code for the Responsible Conduct of Research.

Until such data is secured, the methodology can be exercised on a synthetic plant that mimics the calibrated statistics.

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## 9. Data Management

All code, simulation configurations, model parameters, and analysis notebooks are version-controlled in a GitHub repository. The repository already holds the controller implementations, the literature review, and the implementation walk-through. Development and data analysis are carried out on a Curtin-managed laptop. The codebase and the generated artefacts (simulation outputs, figures, trained model/estimator parameters) are backed up to the Curtin OneDrive area. Any confidential industrial data will be held separately under access control per the data-sharing agreement. It will be stored encrypted, decrypted only during data processing, and never committed to the repository. Research data will be retained for a minimum of seven years in compliance with the Australian Code for the Responsible Conduct of Research.

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## 10. Resources and Budget

The work is primarily computational. The core toolchain is `do-mpc`/CasADi with the IPOPT NLP solver, plus MATLAB/Simulink for cross-validation. It runs on the existing laptop, and the software is free or covered by Curtin licensing, so the current phases carry no cost. Two future needs may require support. The PINN phase may need cloud compute or a GPU workstation for training. Conference attendance and publication charges may also arise, as the first paper will be submitted soon.

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## 11. Conclusion

Comminution dominates a concentrator’s electricity bill. A SAG circuit spends most of its operating life in transients, not at the strict steady state that classical real-time optimisation assumes. The natural control objective is therefore economic. At the SAG-mill level it reduces to minimising specific energy at the demanded throughput and quality.

The candidature rests on two contributions. The first is a modular, energy-minimising cost function that confines the downstream economics to a swappable offline interface. Retargeting the controller to a new ore body or plant then means replacing the specification, not re-engineering the controller. The second is a transient-rich benchmark of the supervisory architectures. The first like-for-like benchmark is complete, and its results support the modular direction. The two-layer EMPC remains the main candidate.

The results suggest a division of labour. Profit optimisation belongs at the plant-wide level, together with planning and scheduling. The circuit level can exploit transients through EMPC with a consistent objective. This division also lowers the barriers that have kept EMPC out of operating plants.

The remaining programme calibrates the model to plant data, builds the plant-representative scenario, and extends the comparison to the full set of system designs: SSRTO, ROPA, single- and two-layer EMPC, and the regulatory baselines. The higher-risk PINN objective is staged so the thesis remains complete regardless of its outcome.

Grinding can draw up to roughly 80 percent of a concentrator’s energy, so even small percentage savings matter at industrial scale. In mining, RTO and EMPC are well established in theory but little used in operating plants due to a number of issues discussed in the report. The end goal is to develop a practical, energy-efficient and robust system design and its implementation approaches that make deployment realistic and solve the main hurdles.

## References

- Bouchard, J., Sbarbaro, D., & Desbiens, A. / Thivierge, A., Bouchard, J., & Desbiens, A. (2023). Comparing economic model predictive control to basic and advanced regulatory control on a simulated high-pressure grinding rolls, ball mill, and flotation circuit. *Journal of Process Control*, 122.
- Chen, W.-H., Yang, J., Guo, L., & Li, S. (2016). Disturbance-observer-based control and related methods - an overview. *IEEE Transactions on Industrial Electronics*.
- Chen, X.-S., Yang, J., Li, S.-H., & Li, Q. (2009). Disturbance observer based multi-variable control of ball mill grinding circuits. *Journal of Process Control*, 19(7).
- Chen, X.-S., Yang, J., Zhong, Z., & Zhai, J. (2021). Process control of ball mill based on MPC-DO.
- Coetzee, L. C., Craig, I. K., & Kerrigan, E. C. (2010). Robust nonlinear model predictive control of a run-of-mine ore milling circuit. *IEEE Transactions on Control Systems Technology*, 18(1).
- Darby, M. L., Nikolaou, M., Jones, J., & Nicholson, D. (2011). RTO: An overview and assessment of current practice. *Journal of Process Control*, 21.
- Ellis, M., Durand, H., & Christofides, P. D. (2014). A tutorial review of economic model predictive control methods. *Journal of Process Control*, 24.
- Ellis, M., Liu, J., & Christofides, P. D. (2017). *Economic Model Predictive Control: Theory, Formulations and Chemical Process Applications*. Springer.
- Faulwasser, T., & Pannocchia, G. (2018). Toward a unifying framework blending real-time optimization and economic model predictive control. *Industrial & Engineering Chemistry Research*.
- Hinde, A. L., & Kalala, J. T. (2009). The application of a simplified approach to modelling tumbling mills, stirred media mills and HPGR's. *Minerals Engineering*, 22(7-8), 633-641.
- Jia, Y., et al. (2020). Multi-stage economic model predictive control for a gold cyanidation leaching process under disturbances. *AIChE Journal*.
- Le Roux, J. D., & Craig, I. K. (2013). Reducing the number of size classes in a cumulative rates model used for process control of a grinding mill circuit. *Powder Technology*, 246, 169-181.

- Le Roux, J. D., & Craig, I. K. (2019). Plant-wide control framework for a grinding mill circuit. *Industrial & Engineering Chemistry Research*, 58.
- Le Roux, J. D., Craig, I. K., Hulbert, D. G., & Hinde, A. L. (2013). Analysis and validation of a run-of-mine ore grinding mill circuit model for process control. *Minerals Engineering*, 43-44, 121-134.
- Le Roux, J. D., Olivier, L. E., Naidoo, M. A., Padhi, R., & Craig, I. K. (2016). Throughput and product quality control for a grinding mill circuit using non-linear MPC. *Journal of Process Control*, 42.
- Le Roux, J. D., Steinboeck, A., Kugi, A., & Craig, I. K. (2017). An EKF observer to estimate semi-autogenous grinding mill hold-ups. *Journal of Process Control*, 51, 27-41.
- Le Roux, J. D., & Steyn, C. W. (2022). Validation of a dynamic non-linear grinding circuit model for process control. *Minerals Engineering*, 187.
- Matias, J., Le Roux, J. D., & Jäschke, J. (2022). Steady-state real-time optimization using transient measurements (experimental rig comparison of SSRT0, ROPA, and DRTO).
- Mittermaier, H. K., Le Roux, J. D., & Craig, I. K. (2025). Model-plant mismatch diagnosis using plant-model ratios for a grinding mill circuit under MPC. *Minerals Engineering*, 227.
- Numbi, B. P., & Xia, X. (2016). Optimal energy control of a crushing process based on vertical shaft impactor. *Applied Energy*, 162.
- Pannocchia, G. (2018). An economic MPC formulation with offset-free asymptotic performance. *IFAC-PapersOnLine*, 51(18).
- Pannocchia, G., Gabiccini, M., & Artoni, A. (2015). Offset-free MPC explained: novelties, subtleties, and applications. *IFAC-PapersOnLine*, 48(23).
- Quintanilla, P., Fernández, F., Mancilla, C., Rojas, M., & Navia, D. (2025). Digital twin with automatic disturbance detection for an expert-controlled SAG mill. *Minerals Engineering*, 220.
- Sosa-Blanco, C., Hodouin, D., Bazin, C., Lara-Valenzuela, C., & Salazar, J. (2000). Economic optimisation of a flotation plant through grinding circuit tuning. *Minerals Engineering*, 13(10-11).